N100 Financial Intelligence Platform — Analyst Guide

Purpose

A read-only research interface for exploring company fundamentals, computed financial ratios, sector and peer context, valuation indicators, cash-flow intelligence, and data-derived company archetypes. The supplied market data is historical and should not be treated as a live quote feed.

API quick start

From the project root: **pip install -r requirements.txt** then **uvicorn src.api.main:app --reload**. Open /docs for interactive Swagger documentation. Set N100_DB_PATH to override the SQLite database path.

Clustering methodology

K-Means uses five features: return_on_equity_pct, debt_to_equity, revenue_cagr_5yr, fcf_cagr_5yr, and operating_profit_margin_pct. Missing values are imputed using sector medians, followed by StandardScaler. The model is fixed at k=5 and random_state=42 for reproducibility. FCF CAGR is derived only where positive endpoints and a sufficient observed time span are available; missing cases use sector median imputation.

Interpretation cautions

Cluster IDs are algorithmic labels and do not imply investment quality or recommendations. Archetype names are descriptive summaries, not ratings. Imputed fields should be considered less certain than observed values. Inspect distance_from_centroid to identify companies less typical of their assigned cluster. Financial metrics inherit source-period limitations and may mix annual/interim observations where flagged in prior sprint reports.

Key outputs

output/cluster_labels.csv, output/elbow_plot.png, output/correlation_heatmap.png, output/outlier_report.csv, output/portfolio_stats.csv, docs/openapi.json, and docs/N100_API.postman_collection.json.